



STAGEMARK 2027

SiLeMi/o · By Mamat · -----[]--

User manual

Metric stage marking, plans, projector calibration and safe output control.

SHOW RULE - The output always opens black. Verify the selected display, then use **SHOW**. **BLACKOUT** remains available and takes priority at all times.

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Windows 11 and macOS · local and offline operation

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This manual follows the recommended path: metric stage → objects → plans → file → calibration → projection → blackout. Persisted coordinates always use integer millimetres.

1. Output safety principles

StageMark prepares and projects vector marks on a stage. It does not control the projector hardware: it uses a secondary display connected to the video system on Windows or macOS, or a Spout 2 source received by mapping software on the same Windows PC.

Three distinct uses: designing the plan needs no projector; making marks visible on the floor needs at least one projector, and sometimes several for a large stage or to reduce shadows; optional XREAL glasses provide a personal AR view that the audience cannot see.

- No projection window exists at startup.
- Opening the output first creates a fully black, borderless surface with no pointer.
- Any display removal or metric change immediately forces blackout.
- The reference background imported into the editor is never sent to the projector.
- BLACKOUT also disarms the local controller's Instant point.
- Spout output must be started explicitly on the PC. It opens black and sends another black frame before stopping.

Before a show: disable notifications, sleep and screen savers; postpone scheduled system restarts; secure the cable path; test BLACKOUT from the computer, phone and Stream Deck.

2. Quick start

The permanent header contains StageFlow connection (LIVE and remote), Outputs & setup (video, AR and calibration), Alerts, the current Light / Dark theme, FR / EN, Guide and Help. Guide opens this StageMark manual in the selected language. Help → SiLeMIO suite guide opens the cross-application document. Secondary controls wrap when space is limited; SHOW and BLACKOUT remain visible.

1. In Stage, enter the real width and depth, then choose the outline and grid.
2. In Drawing, create marks and use numeric fields for exact positions and dimensions.
3. Create or choose the working Plan, then save a standalone StageMark .stagemark project, open a local StageFlow .stageflow project, or explicitly join a StageFlow Network LIVE session.
4. In Outputs & setup → Projector, choose and open the display, or start Spout on Windows. Every output stays black.
5. In Outputs & setup → Setup & calibration, choose the projector, then show the test pattern and align its four corners with the real target.
6. Return to the plan and select SHOW. Use BLACKOUT whenever no image should be visible.

SHOW is disabled until the output is open. BLACKOUT is always available from every workspace.

3. Metric stage and navigation

The origin (0, 0) is at centre stage on the downstage edge. Negative X is stage right/Jardin, positive X is stage left/Cour, and positive Y points upstage. Metres are only a display conversion: project data remains in integer millimetres.

- Project metadata (name, venue, author, event date and notes) is edited in Stage and feeds team outputs.
- The display-unit selector only changes on-screen readings; stored coordinates remain integer millimetres.

Action	Gesture
Zoom	Mouse wheel
Pan the view	Ctrl + drag the background
Alternative pan	Alt + drag or middle mouse button
Precise move	Arrows: 10 mm · Shift + arrows: 100 mm
Add to selection	Ctrl + click · Shift + marquee

Scaled reference plan

- Import a PNG, JPEG, WebP or SVG: the image is embedded in the project and remains strictly editor-only.
- Unlock it and choose Edit on stage. Drag the image to move it; drag a side for a free dimension or a corner to preserve proportions.
- Choose Scale by measure, click two points whose distance is known, enter the real distance in metres and apply. StageMark scales proportionally around the first point, using integer millimetres.

4. Objects, free shapes and plans

Objects belong to the active plan. The permanent Site / obstacles layer is shared by all plans and excluded from projection by default. Edit groups move objects together; display snapshots only store their visibility.

- Create, rename, lock and hide layers, then reassign objects without changing their geometry.
- Distribute a selection along X or Y for regular spacing; existing alignment tools remain available.
- Dimension mode measures a stage distance and can keep a non-projected technical annotation.

Free shapes and Pen tool

- Click: create a corner and a straight segment.
- Click-drag: create a smooth point and Bézier handles.
- Straight and curved gestures can be mixed within one path.
- Enter closes a shape; Shift + Enter finishes an open line; Backspace removes the last point.
- After creation, Edit path can add, remove or convert a point between Corner / straight and Smooth curve.

Plans and visibility

- Create, duplicate, rename, reorder, hide or delete a plan in the left column.
- Create a snapshot from a selection, then Load, Close or Solo it.
- Obstacles + plan or Obstacles only must be selected explicitly during preparation.

Do not use the reference background as a stage mark. It is deliberately removed from every projection payload.

5. Standalone, local StageFlow and Network LIVE projects

One show, three explicit access paths

StageMark	STANDALONE STAGEMARK PROJECT Name.stagemark	No StageFlow required
Suite SiLeMIO	LOCAL STAGEFLOW PROJECT Name.stageflow	Folder accessible on this computer
StageFlow free and optional	STAGEFLOW NETWORK LIVE SESSION	LAN discovery or private IPv4 + 6-digit code

StageMark distinguishes three access paths. A .stagemark file is the complete standalone StageMark project. A .stageflow folder accessible on this computer is a local StageFlow project that StageMark can create, open and save without running StageFlow. A StageFlow Network LIVE session is broadcast by StageFlow on the local network, discovered in StageMark, then joined voluntarily with its 6-digit code. StageFlow is free and optional.

Network LIVE does not share the host computer's path and uses neither cloud storage nor a fake local file. StageMark validates the remote project and assets in memory before enabling following or commands. Source labels follow validated data; your custom text takes priority. Other Patch/CAD changes wait for explicit application. A data update can never show an image, remove BLACKOUT, change the projection display or recall a cue.

- In a standalone .stagemark project, Save writes the complete V11 model to the same file. Save As creates a new .stagemark file and leaves the source untouched.
- In a local .stageflow project, Save writes only stagemark/stagemark.json. Locks, the base hash, manifest merging and atomic replacement protect every foreign domain, including Patch, CAD, StageDesk, StageMon and Dante Config Editor.
- Open StageFlow connection (header button or Ctrl+L). Local project creates or opens .stagemark / .stageflow without LIVE. For a session on this PC or the local network, choose StageFlow LIVE, Refresh sessions, the project and host computer, then enter the 6-digit code. Leading zeroes are preserved; ASCII spaces and hyphens are accepted, but letters and punctuation are never silently repaired. If multicast discovery fails, enter the StageFlow computer's private IPv4 address as a fallback. Pairing is never automatic.
- An available session has not yet been joined. The centre distinguishes active local LIVE, network LIVE and the StageMark remote. If no session appears, first check that StageFlow is open and its LIVE is started. Back to project closes the centre without disconnecting the current work.
- The network project stays in memory only: no cache, autosave or recovery copy is written to disk. The manual address, code and token are not persisted, and StageMark never reconnects by itself.
- Saving in Network LIVE publishes only the StageMark domain with its base hash. A stale base creates a visible conflict and preserves local edits instead of overwriting the remote project.
- Following validates identity, revision and hash. In followed local LIVE or a joined network LIVE session, names and labels still linked to their source update without moving marks. Locked objects or layers stay unchanged. Outside LIVE, use Refresh then Apply in editor.
- Network following and commands start only after the exact project is installed. A different session or project identity is rejected. Explicit show-control commands remain separate from project updates.
- Detaching from Network LIVE, losing the session or failing during activation keeps the drawing as a modified, unsaved standalone StageMark copy.

- The AutoCAD connector for Windows works with a local StageFlow project and can directly join a remote LIVE session. The connector is free and optional; AutoCAD keeps its separate Autodesk license.
- Alerts are ON at every join: an orange operator banner shows previous → current value, label/group, author and time. Acknowledge or Acknowledge all on this workstation targets alerts visible at the click; later changes remain pending. View list opens the details.
- Local reception OFF has no backlog when enabled again. StageFlow's notification pause is shown separately: LIVE, synchronisation and safety continue. Neither setting remotely acknowledges alerts belonging to another workstation.
- Save and Save As always preserve the active project type. Copy commands between .stagemark and .stageflow are explicit and non-destructive, and leave the source untouched.
- Import legacy... opens an old .json file without changing it; its first save creates a .stagemark file. An existing .stagemark file opens directly as a standalone project.
- CadOverlay remains separate from the reference image and its projection is off by default. An update accepted in the editor is never sent to an output without an operator action.
- Local autosave allows recovery on the next launch after a local-project interruption; it is disabled for a Network LIVE session.
- Recent projects reopens a .stagemark file, a .stageflow folder or an imported legacy JSON, removes a missing entry or clears only the history. No project is deleted.
- Keep a backup of your projects and calibrations outside the installation folder before maintenance or moving computers.
- Ctrl+S saves; Ctrl+Shift+S chooses a new local destination of the same type; Ctrl+O offers a standalone project, a local StageFlow project or a StageFlow Network LIVE session.

PDF, PNG and printing

- Choose one or more plans, then inspect the preview before producing any output.
- Enable grid, site/obstacles, reference background, legend, scale and dimensions independently for the team document.
- PDF and printing can create multiple A4 pages, one plan per page. The suggested orientation can always be overridden.
- PNG exports one plan. The filename includes the project, plan and timestamp.

6. Installation, calibration, projection and cueing

Outputs & setup groups settings; close it to return to drawing or Live. Live keeps plans, cues, SHOW / BLACKOUT and independent AR controls; expand its preflight when needed. Its Projector, Placement and Calibration tabs bring together the equipment bank, lenses, calculations and geometry settings.

Use	Required equipment
Plan design	Computer only
Marks visible on the floor	One or more projectors connected to the video system
Personal AR view	ARCore Android or XREAL Air 2 Ultra + Beam Pro; no audience display

- The Technical summary shows the profile, display, resolution and actual on-site validation status. Check calibration, Choose display and output and Configure remote control only navigate; they never enable output.

1. Connect the projector and select an extended display in Windows or macOS settings, at native resolution.
2. In Outputs & setup → Projector, explicitly select the projector display.
3. Select Open output. The chosen display turns black.
4. In Outputs & setup → Setup & calibration, verify the projector and lens, then select Show test pattern in Calibration.
5. Move the four numbered corners. Arrows: 1 px; Shift + arrows: 10 px.
6. When the pattern matches the real target, use SHOW for the plan and BLACKOUT to cut it.

Escape a black screen: Ctrl+Shift+F12 (Cmd+Shift+F12 on Mac), Escape in output, or CLOSE OUTPUT at the top. Closes video, black curtain and Spout without quitting the project or stopping AR. BLACKOUT instead keeps the black window open. Closing output may expose the desktop on the projector.

Calibration profiles

- Create, rename and duplicate a profile for each real installation; a duplicate must be validated again on site.
- Associate the profile with the projection display, projector and lens in use. A technical change automatically invalidates the previous validation.
- Select Validated on site only after physically checking the pattern and blackout. The final profile cannot be deleted.

Calibration corrects reasonable perspective, but cannot replace suitable optics. Strong side or oblique projection reduces focus, brightness and usable resolution.

Optional augmented reality

StageMark AR 0.1.1 build 12 offers an ARCore Android camera view (including Galaxy A54 5G) or XREAL Air 2 Ultra + Beam Pro. Outputs & setup → Augmented reality offers Start AR directly and an optional guide: equipment, measured placement, pairing code and validation. On the phone, allow camera access, scan the floor, capture A/B/C, then tap VALIDER ET AFFICHER LES REPÈRES (validate and show marks). Calculation is automatic. On failure, compare measured/expected distances and redo A, B or C. For a home trial, use stage dimensions actually measured in the room. AR guide.

- SHOW AR / HIDE AR on the PC are independent of video: video only, AR only, both or neither. Start AR opens the session and authorizes marks after pairing and calibration, without a second required button. Show AR / Hide AR remain available throughout. The companion can never remove video BLACKOUT.
- MASQUER POUR MOI (hide for me) in the APK only affects that person and cannot override a PC hide. Hiding keeps camera, connection and calibration while tracking remains valid. Stopping the AR session disconnects companions. An empty plan is identified.

- A/B/C: one reusable set, printed at 100% (full 160 mm image including white margin). Outputs & setup → Augmented reality → A/B/C placement gives coordinates and a project-specific top-view PDF. Reprint freely; recalibrate at each installation.
- Show A/B/C on the plan and Include A/B/C in projection are independent and off by default. Position crosses, not trackable patterns; SHOW/BLACKOUT keep priority. Hide after placement and verify distances with a tape measure.
- Initial Google AR services installation may need Internet. Operation then uses no cloud or camera image upload. Glasses recognise printed 160 mm images; phones capture their centres manually.
- Tracking loss, pause or geometry change: recalibrate. Lighting and a textured floor are needed. Accuracy and drift still require device measurements; not suitable for critical positioning.

Obstacle modes

Mode	Projected content
Off	Active plan objects only
Obstacles + plan	Active plan and permanent Site objects
Obstacles only	Permanent Site objects, with the plan emptied

Local Spout output (Windows only)

For MadMapper, OBS or another Spout receiver on the same PC, choose the sender name, resolution and frame rate, then select Start in BLACKOUT. SHOW, BLACKOUT, the test pattern, plans and Instant point control the direct display, Spout, or both. The first version uses a BGRA CPU copy; check motion and blackout latency at the production resolution.

Cueing and preflight

- Capture cues from the current state, rename and reorder them, then recall them during operation.
- Preflight checks the essentials before SHOW: output, display, calibration, plan and safety state.
- Recalling a cue never bypasses priority blackout. Prepare and rehearse the complete cue list before the audience enters.

7. Local phone controller

In StageFlow connection → StageMark remote, choose the network and explicitly start the controller. Scan the integrated QR or copy its private link to a device on the same network. On the phone, tap Take control: connecting alone does not allow editing. Only one pilot can control it; other devices remain observers. This remote does not require StageFlow.

During a StageFlow Network LIVE session, the optional StageFlow QR portal can open this exact same remote. StageMark must have started the local controller first; the handoff is temporary, does not disclose the primary token and never triggers output. StageMark's direct QR remains available without StageFlow. When StageFlow and StageMark use different IP addresses, the portal opens the remote full-screen; browser Back returns to StageFlow. Opening a connection screen never starts a server or output.

- The phone can open direct display output, close all active outputs, show the plan, engage blackout and show the test pattern.
- Through the StageFlow portal, an authorised connected session stays valid through its heartbeat, instead of disconnecting after five minutes of use. After five minutes without a heartbeat, controller stop, LIVE loss or revocation, access cannot renew itself. Return to the portal for fresh access, then explicitly take control again.
- The Calibration section in Installation adjusts all four corners from the real target surface.
- Instant point projects a temporary mark without changing the file. Blackout removes it.

Create and edit marks

- In the mobile editor, check the target plan, choose a native type and enter its name. Place by touch prepares the next tap on the drawing; Add at center creates the mark at the centre of the stage.
- Select an object on the drawing or in the name list, then edit its name and measurements in the inspector. Positions and dimensions use the same whole-millimetre native model as the PC.
- Choose the move gesture to drag the selected object; choose the scroll/zoom gesture to navigate the page. One continuous move is grouped into one native history step.
- For a free shape, place several vertices, then finish open or closed. Segments stay straight; edit advanced Pen and Bézier handles on the PC.
- Deletion needs confirmation and only targets the selected object. Locked objects or layers stay protected; a plan change, concurrent edit or disconnection cancels a stale gesture instead of applying it elsewhere.
- Creating or moving a mark never starts output or removes BLACKOUT. Remote availability, LIVE connection, pilot rights and output state are four separate states.
- The phone Live tab can choose a plan, recall a cue and inspect preflight while standing on stage.
- After a connection loss, the phone shows OFFLINE / STATE UNKNOWN and disables commands. Nothing is replayed: tap Reconnect, then take control again. A stopped session requires its new QR code. Pending confirmations are cancelled.

The controller uses no cloud or account. It is powerful: share the session QR only with the authorised operator and stop the session after use.

8. Projector placement calculator

The placement calculator sits in Installation, with the projector bank and immediately before calibration. It handles a projector centred above the stage, at front of house, or outside either stage side.

Useful width = $\max(\text{stage width, depth} \times \text{image aspect ratio}) \times \text{margin}$.

Lens-to-floor height = $\text{useful width} \times \text{throw ratio}$.

Input	Source
Resolution	Projector native resolution, for example 1920 × 1200
Minimum / maximum throw	Manufacturer data for the lens actually fitted
Margin	Usually 5 to 15% around the stage

The computed height is lens-to-floor, not rigging-point height. Physically confirm zoom, focus, brightness, shadows, structural load and safe suspension.

For an oblique position, StageMark also estimates centre throw distance, effective throw ratio, angle and perspective spread. Homography cannot restore lost pixels, brightness or depth of field; the final decision belongs to the video and rigging professionals.

The bundled catalog contains ten professional projectors and 100 lenses. Search, filters, sorting and local favourites simplify selection. Applying a record copies only resolution and throw range into the calculator. Catalog V2 verifies its manifest and SHA-256 before import. A photo is shown only when its licence explicitly permits redistribution; no unlicensed manufacturer image is bundled.

9. Stream Deck and shortcuts

Key	Action
F13	Priority BLACKOUT
F14	SHOW the plan
F15 / F16	Previous / next plan
F17 / F18	Open / close output
F19	Show test pattern
F20	Arm Instant point
F21 / F22	Previous / next cue
Ctrl/Cmd+Alt+1 to 8	Recall cues 1 to 8 directly

In Stream Deck, assign the Hotkey action to each key. No StageMark plugin, account or cloud service is required.

OSC and MIDI controls

Open Tools → OSC · MIDI · Stream Deck to configure OSC input and feedback, select MIDI ports, learn a message and open the Stream Deck kit. StageMark stays standalone; in LIVE, StageFlow relays the same identifiers.

Use a press followed by a release for a MIDI button: zero CC values and Note Off do not trigger an action. Changing or disconnecting the input stops its control; restart MIDI explicitly. Cue selection accepts its ID, exact name or one-based position. For MIDI, Note / Program Change number 0 maps to position 1; CC uses its value from 1 to 127. A refusal is displayed without simulating success. OSC/UDP alone does not guarantee packet delivery or uniqueness; StageMark does not automatically resend commands.

ID	OSC address
stagemark.output.show	/silemio/stagemark/output/show
stagemark.output.blackout	/silemio/stagemark/output/blackout
stagemark.cue.select	/silemio/stagemark/cue/select
stagemark.cue.next	/silemio/stagemark/cue/next
stagemark.cue.previous	/silemio/stagemark/cue/previous

10. Manual, license and updates

Guide opens this PDF in English; selecting FR opens the French manual. Help → SiLeMIO suite guide explains cross-application work. Trial and license displays the local status and accepts a signed license code.

Trial and permanent license

- The first 30 days remain fully usable with no reminder.
- After 30 days, every feature remains available. Only a dismissible reminder is shown.
- The permanent license is a one-time €49 including tax purchase. It removes the reminder and remains valid on Windows, macOS and compatible updates.
- Purchase opens explicitly in the browser. Existing V1 licenses remain portable and verified offline. A V2 license needs a first online activation for each installation, then works offline with its signed receipt.
- Commercial V2 licenses include three installations by default. Trial and license shows the count and lets you deactivate this computer. No StageMark account, recurring online check or telemetry is required.

Licensing has no access to the stage, files, calibration, projection, local controller or blackout. A licensing error never blocks operation.

Platforms and equipment

This manual accompanies StageMark 2027.0.6 on Windows and macOS.

Use Windows 11 x64. On Mac, choose the Intel (x64) or Apple Silicon (arm64) DMG matching your computer; the declared minimum is macOS 12. Check the version on the download page: available features can differ between Windows and Mac.

Both platforms offer editing, local projects, network LIVE, the web remote and direct display output. Spout 2 and the local presence/pipe console remain Windows-only. Mac packages have no Apple Developer ID signature or notarization; do not disable macOS protections.

Software checks do not replace rehearsal with real equipment: two computers on the LAN, phone, projector, blackout latency and printer need on-site validation. Use Ctrl on Windows or Cmd on macOS for file commands and the connection center. Drawing gestures explicitly labelled Ctrl still use Ctrl.

Safe updates

At startup, StageMark may find the newest stable version matching this computer's exact OS and architecture among public GitHub Releases, without an account or embedded token. The global Latest label no longer decides the offered version on its own. Releases without a matching installer and SHA-256 asset are ignored; older versions are not automatically installed. Download requires your approval, the installer must match its SHA-256 file, and launching it requires a second explicit action. Launching forces blackout, closes output and stops the local controller.

Explicitly download and run the installer after saving and closing the app; verify the SHA-256 published with the package.

An update is never installed silently and must never be launched during a show.

Quick troubleshooting

Symptom	Check
Output stays black	Is output open? Is SHOW active? Is the correct display selected?
Wrong geometry	Show the pattern again, restore corner order and avoid crossing the quadrilateral.
Phone unavailable	Same private network, session started, current QR, connection allowed by the system firewall.
Update unavailable	Check Internet, then open the public Release manually if necessary.
Installer rejected	Download again: StageMark never accepts a missing or mismatched SHA-256.
License code rejected	Copy the complete code again. Validation is local and the app remains usable.

Practical workflows

Understand a label change

What changes	Result in StageMark
Source Kick becomes Kick IN	Inherited name and text become Kick IN. Position, dimensions and cue do not change.
You named the mark Centre drums	That name stays Centre drums. Text still linked to the source can continue to follow independently.
Notifications paused or alerts acknowledged	Data continues to follow. Acknowledgement only records that you have seen the alert.
Patch adds an object or moves CAD	Review and apply the revision in the editor. Current output stays unchanged until an operator action.

Group variants, not soundcheck

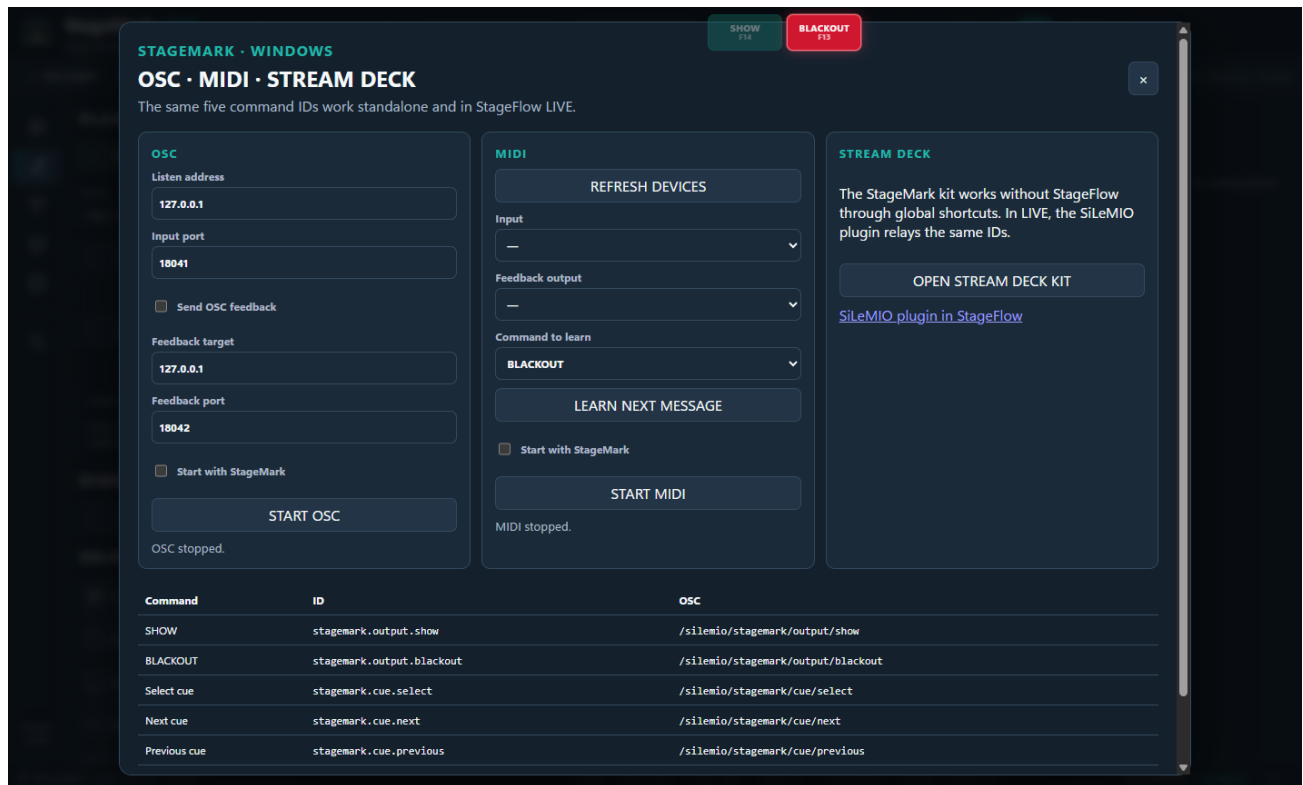
The GENERAL PatchSet supplies shared entries. A group PatchSet can inherit GENERAL and define a specific value for an entry, for example a D6 microphone instead of another model. The technical term override means that specific value, not a soundcheck. Choose the right PatchSet before placing marks. A group entry keeps its identity; renaming GENERAL must not overwrite its custom label. A PatchSet-specific position does not erase the general position.

Prepare a mark from the stage

1. Save the project, leave BLACKOUT active and start the remote on the selected private network.
2. Open the direct QR or StageFlow access, then Take control. Check the project name and target plan.
3. Create a cross named Lead vocal, place it by touch, then adjust its numeric coordinates. Dragging moves the same object; it does not make a copy.
4. Review the mark on the PC and save. To see it on stage, open the output and deliberately use SHOW; finish with BLACKOUT.

On conflict, do not repeat clicks or save attempts. Keep a standalone copy, review the remote revision, then choose which changes to retain.

Connect a show-control command



Tools → OSC · MIDI · Stream Deck: three access paths to common commands, without implicit output startup.

1. Create two different cues, Intro and Final, linked to two plans. Keep output in BLACKOUT during preparation.
2. For MIDI, select the input, choose the command and Learn next message. For OSC, configure address and port before starting input. Software on the same PC can use 127.0.0.1.
3. Send Next cue, Previous cue, then Select cue with its ID, exact name or position. Check the active plan and displayed cue, not only the sent message.
4. SHOW must be refused when no output is open. Only open an output and verify SHOW/BLACKOUT during a controlled equipment rehearsal.

In StageFlow LIVE, explicitly select the StageMark instance attached to the correct show. A lost session, unavailable command or denied permission must never be treated as success. Pending commands are not replayed after reconnection.

License and help

StageMark remains fully usable without a license. The one-time €49 including tax license removes the reminder after the 30-day trial. It is personal, has no subscription, and remains valid through compatible updates. V1 licenses remain portable; V2 licenses use installation activations, which can be released in Trial and license.

Buy a license

<https://stagemark-license.mamat79-dce.workers.dev/buy>

Stripe processes payment in the browser. StageMark does not receive, store or transmit bank details. The license code is then verified offline.

SiLeMi/o suite

StageFlow and its AutoCAD connector are free and optional, with no SiLeMIO activation. StageMark works independently with .stagemark, opens .stageflow locally without StageFlow, and voluntarily joins LAN LIVE with a 6-digit code, without a shared path or cloud. The local console (presence and pipes) is Windows-only, separate from network LIVE and the remote controller. AutoCAD keeps its separate Autodesk license.

StageFlow · StageDesk · StageMark · StageMon · Dante Config Editor · AutoCAD

Website, guides and downloads

<https://www.silemio.com/en/software/stagemark>

StageMark guides · Downloads

<https://github.com/Mamat79/StageMark>

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Thank you for using StageMark. Always run a complete rehearsal with the projector, projection display, remote controller and safety controls before the audience enters.